

Kaung Khant Hein

Full-Stack & Mobile Engineer

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PROFILE

Full Stack & Mobile Engineer proficient in Go, Node.js, PostgreSQL, TypeScript, React 18, and React Native. Proven ability to own software end to end, from multi-tenant backend infrastructure to production client deployment. Blends strong system design with real-world customer ops experience to build fast, scalable applications.

WORK EXPERIENCE

Frontend Web Developer

02/2025 – 08/2025

DevelopX

- Built the core frontend with React 18 and Vite, keeping builds fast and rendering optimized under heavy component trees
- Integrated real-time location discovery with Google Maps API and Leaflet.js, handling geocoding and spatial filtering on the client
- Shipped full i18n across EN, JA, KO, and ZH with i18next, including locale-aware routing, layout reflow, and alignment adjustments
- Designed tab-based navigation with Framer Motion transitions and a favorites system with persistent local state
- Managed global and async state with Redux Toolkit and React Query, keeping API cache invalidation predictable
- Delivered a responsive, mobile-first UI with PWA support and custom Tailwind component primitives

Online Customer Service & E-Commerce Data Entry

01/2024 – 10/2024

Anycall Mobile

- Handled customer support across Facebook, Messenger, and phone channels, resolving inquiries and escalations directly with customers
- Managed product data entry and inventory accuracy using Excel spreadsheets and WordPress plugins for the e-commerce catalog
- Monitored product data quality across listings, performing regular corrections on stock levels, pricing, and product descriptions
- Produced daily and weekly KPI reports tracking support response times, order accuracy, and data entry throughput
- Trained a junior staff member on Messenger automation workflows and order tagging procedures

EDUCATION

Computer Science (Year 1)

University of Computer Studies Mandalay
2019 – 2020

English Language Certificate — B1+ Upper Intermediate

The Space Language Center

Veteran Tour Guide App

A multilingual Southeast Asia tourist web app with real-time location discovery, token-based auth, and four-language support managed through Tolgee cloud.

- › Integrated Google Maps API and Leaflet.js for real-time location discovery with geocoding and spatial filtering
- › Implemented full i18n with Tolgee cloud platform for EN, ZH, JA, and KO, including locale-aware routing and layout reflow
- › Built authentication with Google OAuth and reCAPTCHA, global state with Redux Toolkit, and async data with React Query

React Redux Toolkit Tolgee i18next Google Maps Leaflet

Mingalar Menu — On-Demand Food Delivery App

An on-demand food delivery app where one Go backend serves multiple restaurant brands. Each brand's data is fully isolated through a custom tenancy middleware that reads an X-Brand-ID header and scopes every query downstream.

- › Designed the backend with Domain-Driven Design: eight bounded contexts (brand, catalog, loyalty, order, payment, store, user) each with handler, service, repository, and DTO layers
- › Built a multi-tenancy middleware that enforces brand-scoped data isolation at the HTTP layer. Every database query is tenant-aware without exception
- › Implemented a state-machine-driven order service with transitions (PENDING to PREPARING to READY to COMPLETED/CANCELLED), transactional item pricing with Decimal.js precision, and status audit logging
- › Set up the full infrastructure: Docker multi-stage builds, Docker Compose with PostgreSQL 16, health checks, and a Makefile with 18+ targets for dev workflows
- › Built shared TypeScript packages across the monorepo: API spec (OpenAPI), Zod validation schemas, and typed React Query hooks, keeping frontend and backend contracts in sync

Go Gin GORM PostgreSQL DDD Multi-Tenant Docker Swagger

Apple Things — Full-Stack E-Commerce

An end-to-end mobile commerce platform with a Medusa v2 backend, RBAC, Cloudinary media storage, and an Expo client.

- › Structured the Expo app with atomic design (atoms, molecules, organisms, features) and a strict data flow: UI to hooks to services to Medusa SDK singleton
- › Implemented offline-first persistence with MMKV dual instances (auth-storage for JWT, app-storage for Zustand state) and TanStack Query with async-storage persister for cache hydration
- › Added i18n with i18next and expo-localization, dark/light/system theming with design tokens, and push notifications triggered from Medusa backend events via Firebase Cloud Messaging and Notifee
- › Deployed backend on Render, database on Supabase, media on Cloudinary, and frontend on Vercel with Docker multi-stage builds

Medusa v2 Expo NativeWind MMKV RBAC Cloudinary Supabase Render

Ear Trainer — Web Audio Studio

I wanted to practice ear training for notes and chords, and learn to tune instruments without relying on a third-party tuner. Most apps that promised this either locked core features behind a subscription or felt cluttered with upsells. After researching the Web Audio API and mapping out the app flow I needed, I built my own. The result is a clean, browser-based studio where you can train intervals, identify notes and chords, and tune instruments, all without paying a fee or downloading anything.

- › Built real-time audio synthesis and analysis using the Web Audio API and Tone.js, handling oscillator scheduling, frequency analysis, and waveform rendering without latency
- › Implemented interval, note, and chord recognition with configurable difficulty levels and multi-instrument toggling
- › Designed interactive theory lessons with instant performance feedback and a responsive waveform visualization layer

React TypeScript Web Audio API Tone.js TailwindCSS

Zawgyi Mythic Run — Multiplayer Browser Game

I grew up playing graphic games on PlayStations and consoles, and I missed that era. When I noticed that almost everyone around me now has a laptop or phone, I had the idea of turning those devices into a shared gaming experience. The laptop becomes the monitor, and each phone becomes a console controller. Players join the same lobby through a QR code and play together in real time. That simple nostalgia drove me to build a 2D/3D side-scrolling multiplayer game with room-based matchmaking and real-time input relay through Socket.IO.

- › Built the multiplayer server with room-based matchmaking, 2-player slot management, and real-time input relay through Socket.IO
- › Implemented 5 game modes, LAN IP detection for local play, and Render cloud deployment with active room tracking and disconnect handling

Babylon.js Socket.IO Real-Time Express

Liquid Pill Tabbar — React Native UI Kit

While building a custom bottom navigation bar in a React Native project, I came across the iOS Liquid UI style and loved it. There were libraries that let you install a native liquid bottom nav on iOS, but nothing existed for Android. I decided to build one that was native API independent, so it could run on any platform. What started as a single tabbar component grew into a small component kit: alert toasts, a liquid-styled search bar, bottom sheets, and alert modals, all supporting light and dark themes. I also added brand color tinting as a prop so developers can match the components to any app theme.

- › Published as a reusable npm package (react-native-liquid-pill-tabbar) with glass-morphism aesthetics, animated transitions, and cross-platform compatibility without native module dependencies
- › Expanded into a component kit with toast, search bar, bottom sheet, and alert modal, all accepting a brand color prop for theme-wide tinting

React Native TypeScript npm Expo

Daisy Doll Storefront

When I started a clothing sale page on TikTok targeting customers in Mandalay, I noticed a pattern. Most shops sell through TikTok, but browsing products on the platform is messy. Very few have a website or app, and even those that do feel complicated. Customers do not want to sign up, log in, or download an app just to look at products. They want to browse quickly and then reach out on chat to order. I built a mobile-first web app with that exact flow in mind: no account needed, simple product browsing, and when a customer picks an item, the order text is pre-filled with all the product data from the API and sent directly to our chat. It removed every unnecessary step between "I want this" and "I ordered this."

- Built a zero-friction product gallery with React Query for async data, deployed on Vercel with rewrite rules proxying to the Medusa backend on Render
- Designed for the specific behavior of Myanmar social commerce: browse, select, then message to order with pre-filled product data

React 19 Vite TailwindCSS v4 Medusa v2 Vercel

Technical Exploration — AI & Knowledge Systems

Studied multi-agent orchestration and RAG architecture by setting up and inspecting two open-source reference systems. Focused on understanding workflow orchestration patterns, retrieval ranking, and citation validation, not building from scratch, but learning how production-grade AI systems are structured.

- Explored Helix, an agent orchestration framework: studied the planner, dev, and verifier workflow loop, merge gates, iteration caps, SSE streaming, and run state persistence patterns
- Explored Primer, a knowledge retrieval system: studied authority-based scoring, freshness decay ranking, reciprocal-rank fusion, permission-aware retrieval, and citation validation
- Gained working familiarity with OpenRouter integration, Vercel AI SDK embeddings, event-driven architectures, and connector registry patterns

Open-source projects studied for architecture patterns and AI system design familiarity.

AI Agents RAG Orchestration SSE OpenRouter Vercel AI SDK

TECHNICAL SKILLS

Mobile	React Native (CLI and Expo), TypeScript, NativeWind, MMKV, Notifee, Firebase Cloud Messaging, TanStack React Query, Zustand, i18next
Frontend	React 18/19, Next.js, Vite, Tailwind CSS, Framer Motion, React Hook Form, Zod, MUI, Radix UI
Backend	Go (Gin, GORM), Node.js, Medusa v2, Express 5, REST APIs, Swagger/OpenAPI
Database	PostgreSQL (Supabase, Neon), SQLite, GORM Auto-migrate, Multi-tenant schemas
Desktop	Tauri 2 (Rust), React, Vite, MUI DataGrid, native macOS (ARM/x64) and Windows builds
AI / Systems	OpenRouter, Vercel AI SDK, Multi-agent orchestration patterns, RAG retrieval, SSE streaming
DevOps	Docker (multi-stage), GitHub Actions CI/CD, AWS CloudFormation, Render, Vercel, Netlify, Expo
Storage	Cloudinary (media), Supabase and Neon (PostgreSQL), Excel and WordPress (product data)
Tools	Git, GitHub, Figma, VS Code, ESLint, pnpm Workspaces, Husky, lint-staged